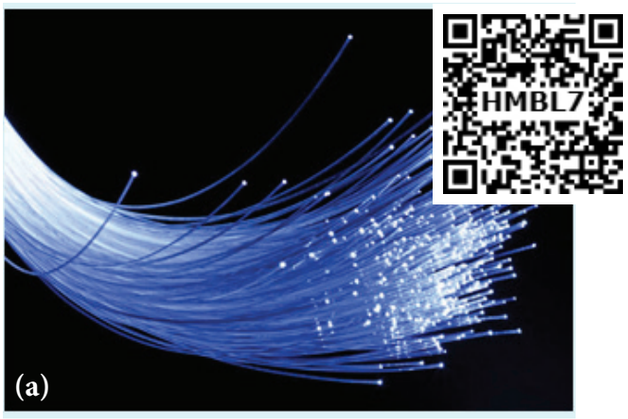


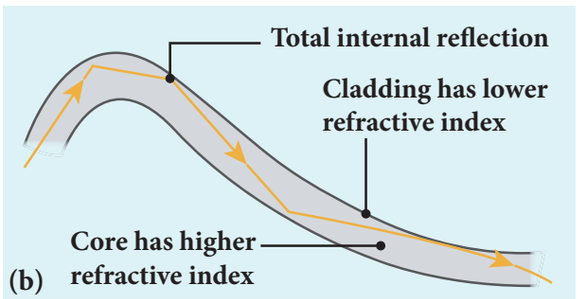


Figure 6.28 Optical fibre

the critical angle. Hence, it advances with repeated total internal reflections inside the optical fibre without undergoing any refraction. The light travels inside the core with no appreciable loss in the intensity of the light as shown in Figure 6.28(a). While bending the optical fibre, it is done in such a way that the condition for total internal reflection is ensured at every reflection inside the fibre as shown in Figure 6.28(b).

6.4.7.6 Acceptance angle in optical fibre

To ensure the critical angle incidence in the core-cladding boundary inside the optical fibre, the light should be incident at a certain angle called *acceptance angle* at the end of the optical fibre while entering into it. It depends on the refractive indices of the core n_1, cladding n_2 and the outer medium n_3. Assume that the light is incident at an angle called acceptance angle i_a at the outer medium-core boundary at A as shown in Figure 6.29(a).

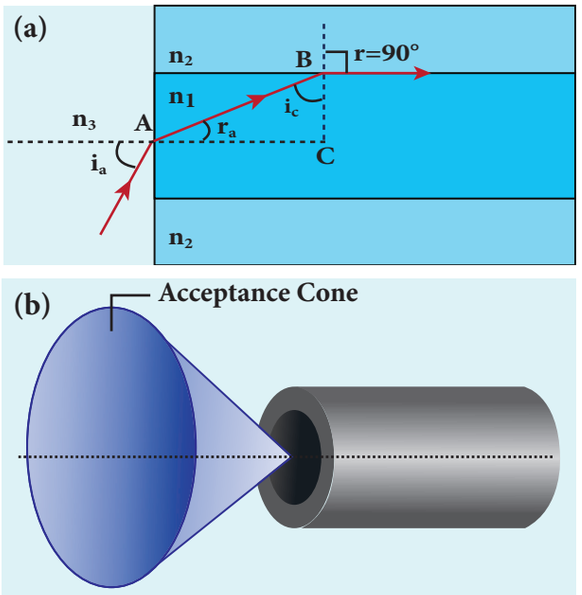


Figure 6.29 (a) acceptance angle and (b) acceptance cone

The Snell's law in the product form, equation (6.19), for this refraction at the point A is,

$$n_3 \sin i_a = n_1 \sin r_a \qquad (6.37)$$

To have the total internal reflection inside the optical fibre, the angle of incidence at the core-cladding interface at B should be atleast critical angle i_c. Snell's law in the product form, equation (6.19), for the refraction at point B is,

$$n_1 \sin i_c = n_2 \sin 90^\circ \qquad (6.38)$$

$$n_1 \sin i_c = n_2 \qquad \because \sin 90^\circ = 1$$

$$\therefore \sin i_c = \frac{n_2}{n_1} \qquad (6.39)$$

From the right angle triangle ΔABC,

$$i_c = 90^\circ - r_a$$

Now, equation (6.39) becomes,

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